

PENETRATION TESTING REPORT

Metasploitable3 Vulnerability Assessment



PENETRATION TESTING REPORT

Metasploitable3 Vulnerability Assessment

Prepared By: Brian Muema
 Classification: CONFIDENTIAL
 Date: May–June 2026
 Scope: Internal Network Penetration Test
 Platform: Kali Linux vs. Ubuntu 14.04 (VirtualBox)

Target System	Metasploitable3 (192.168.56.105)
Prepared By	Brian Muema
Classification	CONFIDENTIAL
Date	May–June 2026
Scope	Internal Network Penetration Test
Platform	Kali Linux vs. Ubuntu 14.04 (VirtualBox)

STRICTLY CONFIDENTIAL — FOR AUTHORISED PERSONNEL ONLY

Table of Contents

Table of Contents	2
1. Executive Summary	3
2. Scope and Objectives	4
2.1 Objectives	4
2.2 Scope	4
3. Methodology	5
4. Reconnaissance	6
4.1 Web Technology Fingerprinting (WhatWeb)	6
4.2 Web Root Directory	6
5. Scanning and Enumeration	7
5.1 Full TCP Port Scan (Nmap -sS -sV -O -A)	7
5.2 Discovered Open Ports Summary	7
5.3 SMB Enumeration — Shares and OS Discovery	8
5.4 SMBClient Share Listing	9
5.5 Directory Brute-Force (Gobuster)	9
6. Vulnerability Analysis	11
6.1 SMB Vulnerability Scan	11
6.2 Metasploit — EternalBlue Attempt	11
6.3 SMB OS and Security Mode Results	12
7. Web Application Findings	14
7.1 Drupal CMS	14
7.2 Chat Application	15
8. Service-Level Findings	16
8.1 FTP — ProFTPD 1.3.5	16
8.2 MySQL — Port 3306	16
9. Risk and Vulnerability Summary	17
10. Open Ports Distribution	18
11. Recommendations	19
12. Conclusion	20

1. Executive Summary

This report presents the findings of a comprehensive penetration test conducted against a Metasploitable3 virtual machine (IP: 192.168.56.105) in a controlled laboratory environment. The assessment was performed by Brian Muema using industry-standard tools and methodologies including Nmap, Metasploit Framework, Gobuster, WhatWeb, and SMBClient.

The target system was found to expose a significant attack surface with multiple critical and high-severity vulnerabilities across several services. The system runs Ubuntu 14.04 (Trusty Tahr) with Samba 4.3.11 masquerading as Windows 6.1, and hosts a variety of outdated and misconfigured network services.

Key findings at a glance:

- 7 open TCP ports identified with exploitable services
- SMB message signing disabled — susceptible to relay attacks
- Anonymous read/write access permitted on IPC\$ share
- Drupal CMS installed with known database warnings and unauthenticated page access
- phpMyAdmin and Payroll application exposed on web root
- ProFTPD 1.3.5 running but anonymous login blocked
- MySQL port 3306 open but remote access restricted to localhost

The overall risk rating of this system is CRITICAL and it is strongly recommended that immediate remediation steps be taken before any production deployment.

2. Scope and Objectives

2.1 Objectives

The objectives of this penetration test were to:

- Identify all open ports and running services on the target machine
- Enumerate operating system information, software versions, and potential vulnerabilities
- Test for SMB-related vulnerabilities using Nmap scripts and Metasploit
- Enumerate web application directories and identify exposed interfaces
- Assess SMB share permissions and anonymous access risks
- Attempt FTP and database connectivity to measure exposure
- Document all findings with supporting evidence (screenshots)

2.2 Scope

Parameter	Detail
Target IP	192.168.56.105
Attacker IP	192.168.56.106 (Kali Linux)
Environment	Oracle VirtualBox — Host-Only Network
Test Type	Grey-box penetration test
Tools Used	Nmap, Metasploit, Gobuster, WhatWeb, SMBClient, MySQL Client, FTP Client
Exclusions	Active exploitation / full compromise beyond enumeration

3. Methodology

The penetration test followed the standard five-phase approach aligned with industry frameworks (PTES / OWASP):

#	Phase	Activities
1	Reconnaissance	Service fingerprinting, OS detection, banner grabbing (Nmap, WhatWeb)
2	Scanning & Enumeration	Full TCP port scan, SMB enumeration, directory brute-force (Gobuster), share listing
3	Vulnerability Analysis	Nmap SMB vuln scripts, Metasploit auxiliary modules, manual web review
4	Exploitation Attempt	EternalBlue module test (unsuccessful — not Windows), SMB version check, FTP/MySQL auth attempts
5	Reporting	Documentation of all findings with screenshots and risk ratings

4. Reconnaissance

4.1 Web Technology Fingerprinting (WhatWeb)

WhatWeb was used for initial web technology fingerprinting, confirming Apache 2.4.7 running on Ubuntu Linux.

```
(kali@kali)-[~]
$ whatweb http://192.168.56.105
http://192.168.56.105 [200 OK] Apache[2.4.7], Country[RESERVED][ZZ], HTTPServer[Ubuntu Linux][Apache/2.4.7 (Ubuntu)], IP[192.168.56.105], Index-Of, Title[Index of /]
```

Figure 1: WhatWeb fingerprint — Apache 2.4.7 on Ubuntu Linux

4.2 Web Root Directory

Browsing to the web root revealed a standard Apache directory listing exposing the chat application, Drupal CMS, phpMyAdmin panel, payroll PHP application, and an uploads directory.

Index of /

<u>Name</u>	<u>Last modified</u>	<u>Size</u>	<u>Description</u>
 chat/	2020-10-29 19:37	-	
 drupal/	2011-07-27 20:17	-	
 payroll_app.php	2020-10-29 19:37	1.7K	
 phpmyadmin/	2013-04-08 12:06	-	

Apache/2.4.7 (Ubuntu) Server at 192.168.56.105 Port 80

Figure 2: Apache index listing — exposed web directories

5. Scanning and Enumeration

5.1 Full TCP Port Scan (Nmap -sS -sV -O -A)

A comprehensive Nmap SYN scan with service version detection and OS fingerprinting was performed. The scan identified 7 open TCP ports.

```

kali@kali: ~
Session Actions Edit View Help

(kali@kali)-[~]
$ nmap -sS -sV -O -A 192.168.56.105 -oN full_scan.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-31 09:00 -0400
Nmap scan report for 192.168.56.105
Host is up (0.0011s latency).
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      ProFTPD 1.3.5
22/tcp    open  ssh      OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|_ 1024 2b:2e:1f:a4:54:26:87:76:12:26:59:58:0d:da:3b:04 (DSA)
|_ 2048 c9:ac:70:ef:f8:de:8b:a3:a3:44:ab:3d:32:0a:5c:6a (RSA)
|_ 256 c0:49:cc:18:7b:27:a4:07:0d:2a:0d:bb:42:4c:36:17 (ECDSA)
|_ 256 a0:76:f3:76:f8:f0:70:4d:09:ca:e1:10:fd:a9:cc:0a (ED25519)
80/tcp    open  http     Apache httpd 2.4.7
|_ http-ls: Volume /
|_  SIZE  TIME      FILENAME
|_  -    2020-10-29 19:37  chat/
|_  -    2011-07-27 20:17  drupal/
|_  1.7K  2020-10-29 19:37  payroll_app.php
|_  -    2013-04-08 12:06  phpmyadmin/
|_ http-title: Index of /
|_ http-server-header: Apache/2.4.7 (Ubuntu)
445/tcp    open  netbios-ssn Samba smbd 4.3.11-Ubuntu (workgroup: WORKGROUP)
631/tcp    open  ipp      CUPS 1.7
|_ http-server-header: CUPS/1.7 IPP/2.1
|_ http-methods:
|_   Potentially risky methods: PUT
|_ http-robots.txt: 1 disallowed entry
|_ /
|_ http-title: Home - CUPS 1.7.2

```

Figure 3: Nmap full scan output — open ports and service versions

5.2 Discovered Open Ports Summary

Port	State	Service	Version / Detail
21/tcp	Open	FTP	ProFTPD 1.3.5
22/tcp	Open	SSH	OpenSSH 6.6.1p1 Ubuntu
80/tcp	Open	HTTP	Apache httpd 2.4.7 (Ubuntu)
445/tcp	Open	SMB / NetBIOS	Samba smbd 4.3.11-Ubuntu
631/tcp	Open	IPP	CUPS 1.7
3306/tcp	Open	MySQL	Unauthorized (localhost only)

8080/tcp	Open	HTTP-Alt	Jetty 8.1.7.v20120910 — 404
----------	------	----------	-----------------------------

5.3 SMB Enumeration — Shares and OS Discovery

Nmap SMB scripts were used to enumerate share permissions and OS information.

```
(kali@kali)-[~]
$ nmap --script smb-enum-shares,smb-os-discovery -p445 192.168.56.105
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-31 22:32 -0400
Nmap scan report for 192.168.56.105
Host is up (0.0017s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:53:B0:B8 (Oracle VirtualBox virtual NIC)

Host script results:
| smb-os-discovery:
|   OS: Windows 6.1 (Samba 4.3.11-Ubuntu)
|   Computer name: metasploitable3-ub1404
|   NetBIOS computer name: METASPLOITABLE3-UB1404\x00
|   Domain name: \x00
|   FQDN: metasploitable3-ub1404
|   System time: 2026-06-01T02:32:18+00:00
|_ smb-enum-shares:
|   account_used: guest
|   \\192.168.56.105\IPC$:
|     Type: STYPE_IPC_HIDDEN
|     Comment: IPC Service (metasploitable3-ub1404 server (Samba, Ubuntu))
|     Users: 1
|     Max Users: <unlimited>
|     Path: C:\tmp
|     Anonymous access: READ/WRITE
|     Current user access: READ/WRITE
|   \\192.168.56.105\print$:
|     Type: STYPE_DISKTREE
|     Comment: Printer Drivers
|     Users: 0
|     Max Users: <unlimited>
|     Path: C:\var\lib\samba\printers
```

Figure 4: Nmap smb-enum-shares and smb-os-discovery results


```

smb-os-discovery:
  OS: Windows 6.1 (Samba 4.3.11-Ubuntu)
  Computer name: metasploitable3-ub1404
  NetBIOS computer name: METASPLOITABLE3-UB1404\x00
  Domain name: \x00
  FQDN: metasploitable3-ub1404
  System time: 2026-06-01T02:32:18+00:00
smb-enum-shares:
  account_used: guest
  \\192.168.56.105\IPC$:
    Type: STYPE_IPC_HIDDEN
    Comment: IPC Service (metasploitable3-ub1404 server (Samba, Ubuntu))
    Users: 1
    Max Users: <unlimited>
    Path: C:\tmp
    Anonymous access: READ/WRITE
    Current user access: READ/WRITE
  \\192.168.56.105\print$:
    Type: STYPE_DISKTREE
    Comment: Printer Drivers
    Users: 0
    Max Users: <unlimited>
    Path: C:\var\lib\samba\printers
    Anonymous access: <none>
    Current user access: <none>
  \\192.168.56.105\public:
    Type: STYPE_DISKTREE
    Comment: WWW
    Users: 0
    Max Users: <unlimited>
    Path: C:\var\www\html\
    Anonymous access: <none>
    Current user access: <none>
Nmap done: 1 IP address (1 host up) scanned in 25.72 seconds

```

Figure 5: Detailed SMB share access — IPC\$ read/write exposed

5.4 SMBClient Share Listing

SMBClient confirmed available shares: print\$, public, and IPC\$, with the latter accessible anonymously.

```

(kali@kali)-[~]
$ smbclient -L //192.168.56.105 -N

      Sharename      Type      Comment
      -----
      print$         Disk      Printer Drivers
      public         Disk      WWW
      IPC$           IPC       IPC Service (metasploitable3-ub1404 server (Samba, Ubuntu))
Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to 192.168.56.105 failed (Error NT_STATUS_IO_TIMEOUT)
Unable to connect with SMB1 -- no workgroup available

(kali@kali)-[~]
$

```

Figure 6: SMBClient listing of available shares

5.5 Directory Brute-Force (Gobuster)

Gobuster with the dirb/common.txt wordlist enumerated accessible web directories.

```
Gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)

[+] Url: http://192.168.56.105
[+] Method: GET
[+] Threads: 10
[+] Wordlist: /usr/share/wordlists/dirb/common.txt
[+] Negative Status codes: 404
[+] User Agent: gobuster/3.8.2
[+] Timeout: 10s

Starting gobuster in directory enumeration mode

.hta (Status: 403) [Size: 285]
.htpasswd (Status: 403) [Size: 290]
.htaccess (Status: 403) [Size: 290]
chat (Status: 301) [Size: 314] [→ http://192.168.56.105/chat/]
cgi-bin/ (Status: 403) [Size: 289]
drupal (Status: 301) [Size: 316] [→ http://192.168.56.105/drupal/]
phpmyadmin (Status: 301) [Size: 320] [→ http://192.168.56.105/phpmyadmin/]
server-status (Status: 403) [Size: 294]
uploads (Status: 301) [Size: 317] [→ http://192.168.56.105/uploads/]
Progress: 4613 / 4613 (100.00%)

Finished
```

Figure 7: Gobuster results — chat, drupal, phpmyadmin, uploads discovered

6. Vulnerability Analysis

6.1 SMB Vulnerability Scan

Nmap's smb-vuln* script suite was executed against port 445. The service was found NOT vulnerable to MS17-010 (EternalBlue) or MS10-054, but a denial-of-service vulnerability in the regsvc service was confirmed.

```
(kali㉿kali)-[~]
$ nmap --script smb-vuln* -p 445 192.168.56.105
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-31 09:18 -0400
Nmap scan report for 192.168.56.105
Host is up (0.00076s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:53:B0:B8 (Oracle VirtualBox virtual NIC)

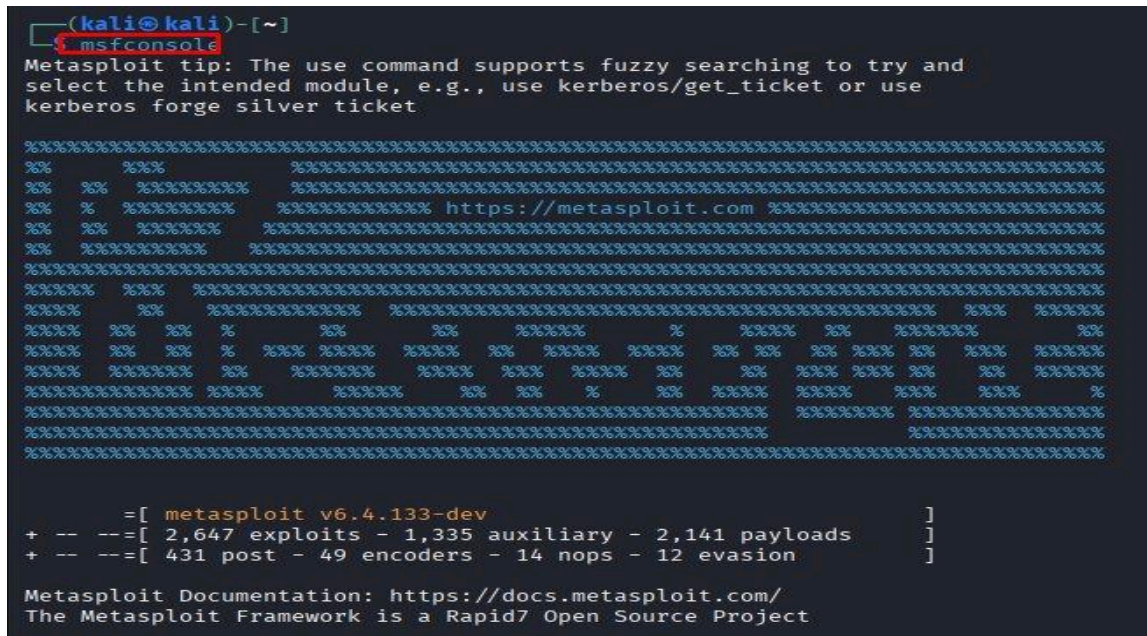
Host script results:
|_smb-vuln-ms10-061: false
|_smb-vuln-regsvc-dos:
|   VULNERABLE:
|     Service regsvc in Microsoft Windows systems vulnerable to denial of service
|     State: VULNERABLE
|       The service regsvc in Microsoft Windows 2000 systems is vulnerable to denial of service caused by a null deference
|       pointer. This script will crash the service if it is vulnerable. This vulnerability was discovered by Ron Bowes
|       while working on smb-enum-sessions.
|_
|_smb-vuln-ms10-054: false

Nmap done: 1 IP address (1 host up) scanned in 40.96 seconds
(kali㉿kali)-[~]
```

Figure 8: Nmap smb-vuln scan — regsvc DoS vulnerability confirmed

6.2 Metasploit — EternalBlue Attempt

The MS17-010 EternalBlue exploit module was attempted in Metasploit Framework v6.4.133. The auxiliary SMB version scanner confirmed the host runs Samba 6.1.0 (not Windows), making EternalBlue inapplicable. SMB signing was confirmed as not required.



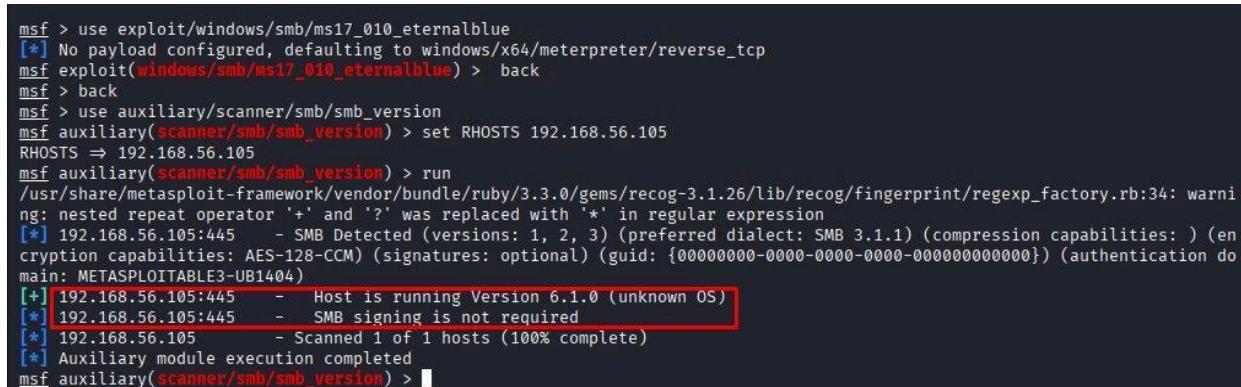
```
(kali@kali)-[~]
msfconsole
Metasploit tip: The use command supports fuzzy searching to try and
select the intended module, e.g., use kerberos/get_ticket or use
kerberos forge silver ticket

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%          %%          %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %  %%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%  %%  %%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

      =[ metasploit v6.4.133-dev                               ]
+ -- --=[ 2,647 exploits - 1,335 auxiliary - 2,141 payloads     ]
+ -- --=[ 431 post - 49 encoders - 14 nops - 12 evasion        ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project
```

Figure 9: Metasploit Framework launched



```
msf > use exploit/windows/smb/ms17_010_eternalblue
[*] No payload configured, defaulting to windows/x64/meterpreter/reverse_tcp
msf exploit(windows/smb/ms17_010_eternalblue) > back
msf > back
msf > use auxiliary/scanner/smb/smb_version
msf auxiliary(scanner/smb/smb_version) > set RHOSTS 192.168.56.105
RHOSTS => 192.168.56.105
msf auxiliary(scanner/smb/smb_version) > run
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/recog-3.1.26/lib/recog/fingerprint/regexp_factory.rb:34: warni
ng: nested repeat operator '+' and '?' was replaced with '*' in regular expression
[*] 192.168.56.105:445 - SMB Detected (versions: 1, 2, 3) (preferred dialect: SMB 3.1.1) (compression capabilities: ) (en
ryption capabilities: AES-128-CCM) (signatures: optional) (guid: {00000000-0000-0000-0000-000000000000}) (authentication do
main: METASPLOITABLE3-UB1404)
[+] 192.168.56.105:445 - Host is running Version 6.1.0 (unknown OS)
[*] 192.168.56.105:445 - SMB signing is not required
[*] 192.168.56.105 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/smb/smb_version) >
```

Figure 10: SMB version scan — Samba 6.1.0, signing not required

6.3 SMB OS and Security Mode Results

Nmap's full scan confirmed message signing is disabled — a dangerous but default configuration enabling NTLM relay attacks — and revealed the SMB dialect supports versions 1, 2, and 3.

```
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: Hosts: 127.0.0.1, METASPLOITABLE3-UB1404; OSs: Unix, Linux; CPE: cpe:/o:lin

Host script results:
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled but not required
| smb-os-discovery:
|   OS: Windows 6.1 (Samba 4.3.11-Ubuntu)
|   Computer name: metasploitable3-ub1404
|   NetBIOS computer name: METASPLOITABLE3-UB1404\x00
|   Domain name: \x00
|   FQDN: metasploitable3-ub1404
|_  System time: 2026-05-31T13:00:57+00:00
| smb2-time:
|   date: 2026-05-31T13:01:00
|_  start_date: N/A

TRACEROUTE
HOP RTT      ADDRESS
1   1.12 ms  192.168.56.105

OS and Service detection performed. Please report any incorrect results at https://nmap.org
Nmap done: 1 IP address (1 host up) scanned in 62.52 seconds

(kali@kali)-[~]
$
```

Figure 11: SMB security mode — signing disabled, guest authentication

```
445/tcp open  netbios-ssn Samba smbd 4.3.11-Ubuntu (workgroup: WORKGROUP)
631/tcp open  ipp          CUPS 1.7
|_ http-server-header: CUPS/1.7 IPP/2.1
|_ http-methods:
|   Potentially risky methods: PUT
|_ http-robots.txt: 1 disallowed entry
|_/
|_ http-title: Home - CUPS 1.7.2
3000/tcp closed ppp
3306/tcp open  mysql       MySQL (unauthorized)
8080/tcp open  http       Jetty 8.1.7.v20120910
|_ http-server-header: Jetty(8.1.7.v20120910)
|_ http-title: Error 404 - Not Found
8181/tcp closed intermapper
MAC Address: 08:00:27:53:B0:B8 (Oracle VirtualBox virtual NIC)
Aggressive OS guesses: Linux 3.2 - 4.14 (98%), Linux 3.8 - 3.16 (98%), Linux 3.10 - 4.11
x 3.13 (94%), Linux 3.13 - 3.16 (94%), OpenWrt Chaos Calmer 15.05 (Linux 3.18) or Designa
, Linux 4.10 (94%), Android 5.0 - 6.0.1 (Linux 3.4) (94%), Android 8 - 9 (Linux 3.18 - 4.
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: Hosts: 127.0.0.1, METASPLOITABLE3-UB1404; OSs: Unix, Linux; CPE: cpe:/o:lin

Host script results:
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled but not required
| smb-os-discovery:
|   OS: Windows 6.1 (Samba 4.3.11-Ubuntu)
```

Figure 12: SMB detailed service results — Samba 4.3.11, WORKGROUP

7. Web Application Findings

7.1 Drupal CMS

A Drupal 7 installation was discovered at <http://192.168.56.105/drupal/>. The site displayed database-related PHP warnings indicating misconfigured or incompatible runtime environments, and an unauthenticated content page was accessible.

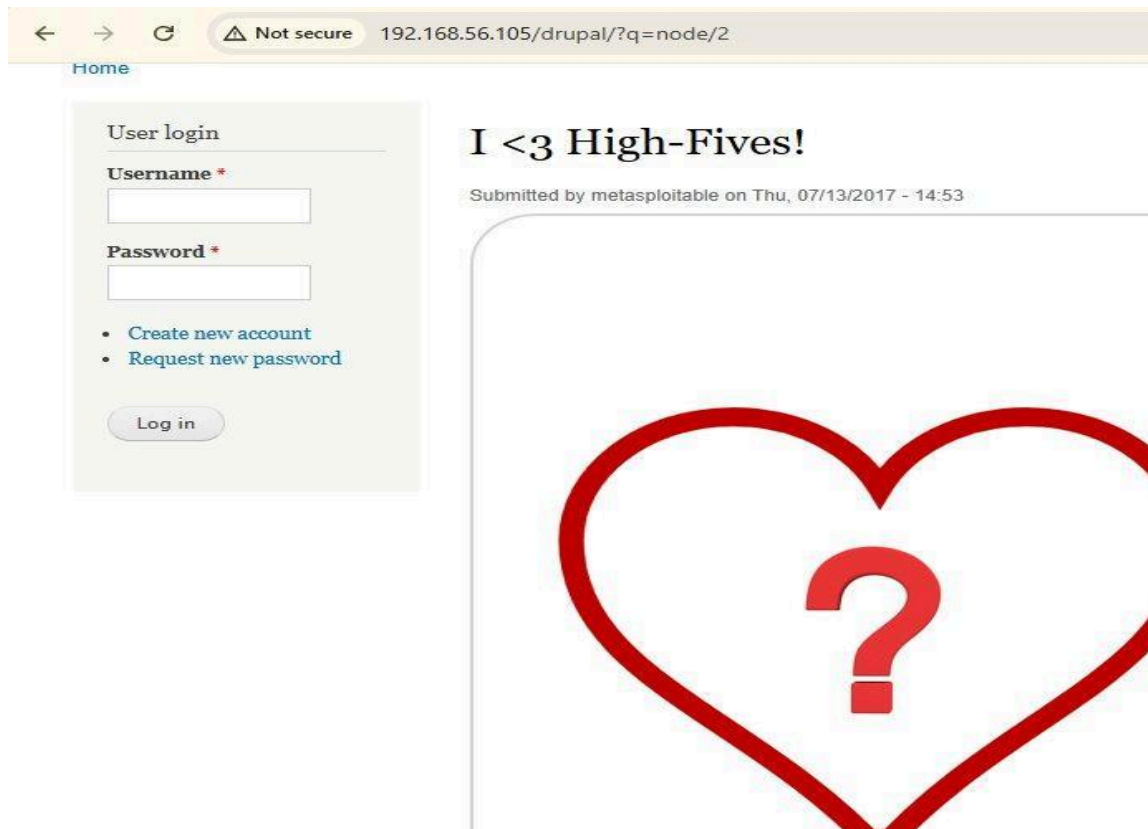


Figure 13: Drupal login page — unauthenticated access to content

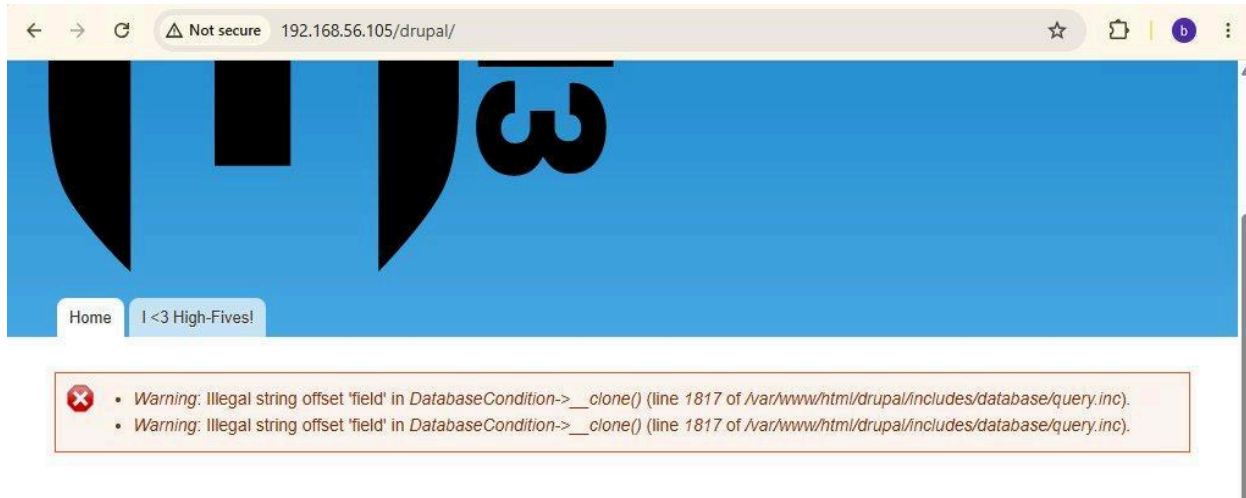


Figure 14: Drupal database error — PHP DatabaseCondition warning

7.2 Chat Application

A web-based chat application was accessible and appeared to expose a user session (Welcome, Brian Muema) without authentication controls, suggesting a stored session or absent authentication.

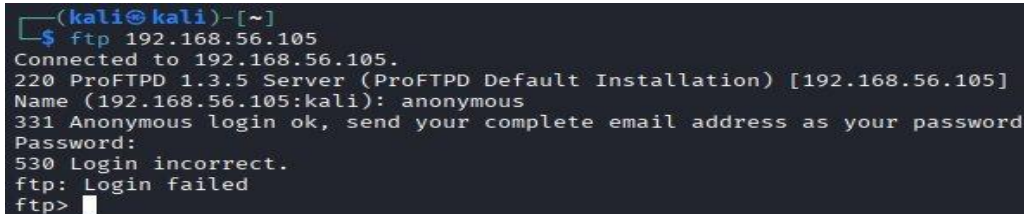


Figure 15: Chat application — logged in as Brian Muema without authentication

8. Service-Level Findings

8.1 FTP — ProFTPD 1.3.5

An FTP connection was established to port 21. Anonymous login was rejected, however ProFTPD 1.3.5 is a known vulnerable version (CVE-2015-3306 — mod_copy arbitrary file read/write without authentication).

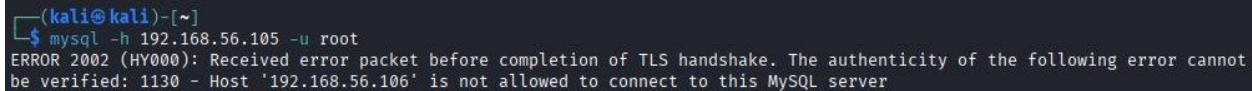


```
(kali@kali)-[~]
$ ftp 192.168.56.105
Connected to 192.168.56.105.
220 ProFTPD 1.3.5 Server (ProFTPD Default Installation) [192.168.56.105]
Name (192.168.56.105:kali): anonymous
331 Anonymous login ok, send your complete email address as your password
Password:
530 Login incorrect.
ftp: Login failed
ftp>
```

Figure 16: FTP connection — anonymous login rejected

8.2 MySQL — Port 3306

A MySQL connection was attempted from the attacking machine. The server rejected the connection with host-based access control: Host 192.168.56.106 is not allowed to connect. While remote root access is blocked, MySQL is running and bound to a reachable port.



```
(kali@kali)-[~]
$ mysql -h 192.168.56.105 -u root
ERROR 2002 (HY000): Received error packet before completion of TLS handshake. The authenticity of the following error cannot be verified: 1130 - Host '192.168.56.106' is not allowed to connect to this MySQL server
```

Figure 17: MySQL connection attempt — host-based denial

9. Risk and Vulnerability Summary

#	Vulnerability	Service	Risk Level	CVE / Ref
1	SMB message signing disabled	SMB/445	HIGH	CWE-294
2	Anonymous IPC\$ read/write access	SMB/445	HIGH	CWE-284
3	RegsvcsDoS vulnerability (SMB)	SMB/445	MEDIUM	smb-vuln-regsvc
4	ProFTPD 1.3.5 mod_copy RCE	FTP/21	CRITICAL	CVE-2015-3306
5	Drupal 7 — outdated CMS, DB errors	HTTP/80	HIGH	SA-CORE-2018
6	phpMyAdmin exposed on web root	HTTP/80	HIGH	CWE-552
7	Chat app — unauthenticated session	HTTP/80	MEDIUM	CWE-306
8	MySQL open to network (port 3306)	MySQL/3306	MEDIUM	CWE-732
9	Jetty server exposed — 404 error	HTTP/8080	LOW	—
10	CUPS 1.7 exposed on port 631	IPP/631	LOW	CVE-2015-1158

10. Open Ports Distribution

The chart below illustrates the distribution of open TCP services identified during the scanning phase. HTTP constitutes the largest share due to its hosting of multiple web applications.

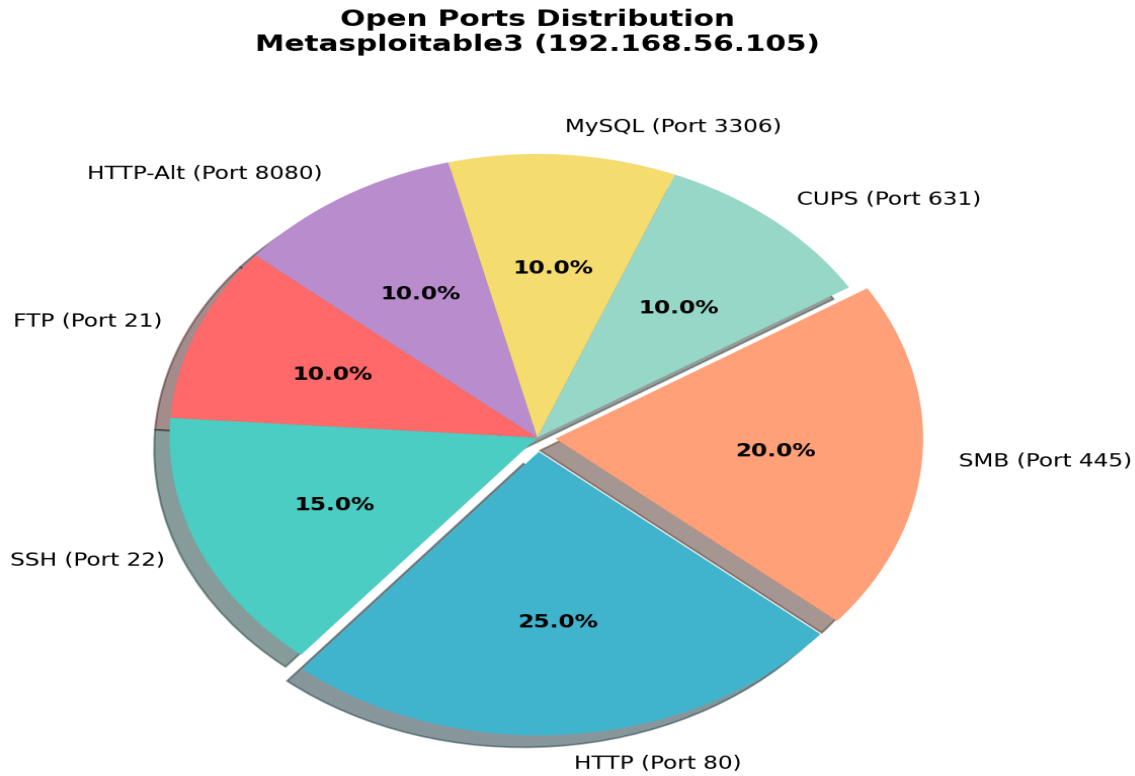


Figure 18: Open ports distribution — Metasploitable3 (192.168.56.105)

11. Recommendations

#	Finding	Priority	Recommended Action
1	ProFTPD 1.3.5 mod_copy RCE	CRITICAL Immediate	Upgrade to ProFTPD 1.3.6+ or replace with vsftpd; disable mod_copy module
2	SMB signing disabled	HIGH — Urgent	Enable SMB signing in smb.conf: server signing = mandatory
3	Anonymous IPC\$ access	HIGH — Urgent	Restrict anonymous access: restrict anonymous = 2 in smb.conf
4	Drupal 7 outdated	HIGH — Urgent	Upgrade to Drupal 10/11 or migrate; apply all security patches
5	phpMyAdmin exposed	HIGH — Urgent	Move behind VPN or firewall; enforce strong authentication and IP restriction
6	MySQL port exposed	MEDIUM	Bind MySQL to localhost only (bind-address = 127.0.0.1 in my.cnf)
7	Chat app — no auth	MEDIUM	Implement session-based authentication; audit all session tokens
8	CUPS 1.7 exposed	LOW	Upgrade CUPS; restrict port 631 to localhost if printing is not required

12. Conclusion

This penetration test of Metasploitable3 (192.168.56.105) revealed a system with a broad and poorly secured attack surface. Multiple services are running outdated software with known CVEs. The SMB service is particularly concerning due to disabled signing and anonymous share access, while the web applications expose administrative interfaces without adequate access controls.

The most critical finding is the presence of ProFTPD 1.3.5, which is vulnerable to CVE-2015-3306 (mod_copy module) — an unauthenticated remote code execution vulnerability. Combined with the exposed phpMyAdmin panel, Drupal 7 installation, and permissive SMB configuration, the system would be trivially compromised in a real-world scenario.

It is strongly recommended that all findings be remediated in order of priority before this system or any similar configuration is deployed in a production network. Regular vulnerability assessments and patch management processes should be established.

Report prepared by: Brian Muema

Penetration Tester | Ethical Hacking & Security Analyst

Date: June 2026 | Environment: Kali Linux / VirtualBox / Metasploitable3